

MULTIMODAL TRANSFORMER-BASED PREDICTION OF BIOMEDICAL TIME SERIES FROM WEARABLE SENSORS: A CASE STUDY IN NEXT-HOUR GLUCOSE FORECASTING

Livia Zaharia^{1,2,3}, Andrii Sameliuk², Alex Karmazin², Nikolay Usanov^{2,3}, Anton Kulaga^{1,2}

¹ IBIMA, Rostock University Medical Center, Germany, livia.zaharia@uni-rostock.de

² GlucoseDAO ³ HEALES (Healthy Life Extension Society)

Background: Wearable devices generate dense, multimodal physiological time series including continuous glucose monitoring (CGM), heart rate, and step counts. These signals arrive at different sampling rates and carry complementary metabolic and cardiovascular information. Recent foundation models for wearable biosignals (Luo et al., 2024; Lutsker et al., 2024) show that transformer architectures with cross-sensor attention can generalize across sensor types and populations. The shared challenge is fusing heterogeneous, irregularly sampled time series into models that transfer between cohorts. A separate gap is evaluation: published models are compared only against other models, with no reference to human performance on the same task.

Methods: We present Sugar I, a transformer architecture for 60-minute blood glucose prediction. Inspired by GluMind (Bian et al., 2025), Sugar I uses parallel cross-attention and multi-scale attention to integrate CGM with heart rate and step count signals at different frequencies. The architecture is not glucose-specific: cross-attention aligns any set of irregularly sampled time series, and multi-scale attention captures both rapid fluctuations and longer-range patterns. We evaluated Sugar I on the AI-READI dataset (896 participants across four cohorts: healthy, pre-T2DM, T2DM oral, T2DM insulin) and on T1DM data from GlucoBench (Sergazinov et al., 2023). To establish a human performance baseline, we developed Sugar-Sugar (github.com/GlucoseDAO/sugar-sugar), a bilingual web application where participants view 3 hours of real CGM history and draw their prediction for the next 60 minutes by placing 12 points on a chart. The Sugar-Sugar study has received ethics committee approval at UMR Rostock and is currently in recruitment, targeting 200+ participants across four groups defined by diabetes status and CGM experience.

Results: On AI-READI (averaged across cohorts), Sugar I achieves MAE 11.39 mg/dL, RMSE 17.62 mg/dL, and MARD 8.25% for 60-minute predictions, compared to 12.95 / 18.19 / n.a. for the best GluMind results and 19.88 / 31.81 / 13.39% for the NHITS baseline. On T1DM data, Sugar I obtains MAE 14.51 and MARD 10.99%, outperforming NHITS (15.11 / 11.24%) and GluFormer (19.53 / 15.10%), though NHITS retains lower RMSE on this cohort (21.05 vs. 23.00). These results use a subset of available data; we have catalogued 53 public CGM datasets with nine device-specific format converters, and training on the full corpus is ongoing.

Conclusions: Sugar I demonstrates that multimodal cross-attention transformers provide a general-purpose architecture for fusing heterogeneous biomedical time series from wearable sensors. The Sugar-Sugar human benchmarking study will produce the first published human accuracy reference for glucose forecasting, and the methodology is transferable to any biomedical time series domain. Code and pipelines are at github.com/GlucoseDAO; model weights on HuggingFace. Supported by a BGI × SingularityNET Benevolent AI grant.

References: Luo, Y. et al. (2024). arXiv:2412.09758. Lutsker, G. et al. (2024). arXiv:2408.11876. Bian, Y. et al. (2025). arXiv:2509.18457. AI-READI Consortium (2024). aireadi.org. Sergazinov, R. et al. (2023). arXiv:2309.01843.

Keywords: biomedical time series, multimodal sensor fusion, transformer, continuous glucose monitoring, wearable health data, human performance benchmarking